RESEARCH & PROTOTYPE REVIEW

SWOT Analysis Report

Source-informed strengths, weaknesses, opportunities, threats and presentation priorities.

SIH 2026 | PS SIH26250 | Alchemy Architects

SOURCE-CODE REVIEW + PUBLIC RESEARCH | 20 OCTOBER 2026

ALCHEMY ARCHITECTS

Executive summary

The SWOT below assesses AAKAASH ADHINAYAK as an early-stage software prototype, based on the provided source export, screenshots and public references. It distinguishes internal engineering assets from unproven outcomes. It is not a market survey or an official evaluation by an SIH organiser or ministry.

1. SWOT matrix

STRENGTHS	WEAKNESSES
- Broad React/TypeScript module structure and coherent dark command-centre design. - Deterministic task/resource scheduling and candidate scoring logic. - Separate rule-based validator and warning states for stale/unknown data. - Scenario replay, audit/evidence concepts and requirements matrix. - Public-data adapters are architected as separate services.	- Live hosted app was not independently smoke-tested during this review. - No verified training/inference code for the displayed RUL metrics. - Hard-coded example metrics and seed telemetry risk being mistaken for measured results. - Validator does not visibly check all expected constraints, including deadline and task-accounting checks. - Source includes operationally themed screens that complicate public safe-demo positioning; original Firestore rules were overly broad.
OPPORTUNITIES	**THREATS**
- Turn the core into a transparent generic logistics/education benchmark. - Use officially documented public civil weather and permitted NASA simulation data. - Publish reproducible benchmarks and negative tests rather than large feature counts. - Conduct authorized consultations with AAI, IMD and training institutions. - Offer education licences or a civilian workflow pilot after user discovery.	- Public release of unauthorised/sensitive datasets or exposed secrets. - Unsupported claims of real-time connectors, fleet readiness or model accuracy. - Confusion between public civil information and authorized defence information. - Solver edge cases or stale data causing false confidence. - No customer demand, procurement path or recurring revenue proof. - Dependency drift and weak rules/authentication in original export.

2. Strategy derived from SWOT

Strategy	Action
SO - use strengths to capture opportunities	Lead with a measurable generic planning sandbox: task import, explicit unknown states, deterministic scheduling, independent checks and evidence export.
WO - repair weaknesses while pursuing opportunities	Replace static predictive metrics with a reproducible benchmark; strengthen validator coverage; prove public connector status through captured evidence.
ST - use strengths to reduce threats	Keep human review, source provenance, versioned scenarios and explicit PASS/REVIEW/REJECT states; avoid automatic action.
WT - limit exposure while repairing fundamentals	Do not make the repository public until source, data, image rights, secrets and Git history have been reviewed; disable or remove operationally themed modules from a public demo.

3. Critical issues to address before SIH presentation

- Replace “ML accuracy improvement” claims with “planned predictive-maintenance experiment” until a run log, split manifest and prediction file reproduce the metrics.
- Replace “optimal solution” language with “candidate from a deterministic heuristic” unless a true solver can prove optimality for the declared model.
- Clearly badge data as LIVE PUBLIC, CACHED, SYNTHETIC or UNKNOWN and expose freshness/observation/retrieval timestamps.
- Do not claim field visits, office endorsement, operational data access or MoD approval without evidence.
- Present only the safe, non-operational planning and verification use case to public evaluators. Avoid tactical, weapon or real-force readiness use cases.
- Make build, unit, import, API and security checks reproducible from a clean clone.

4. Decision

Retain the project direction, reduce the public scope to the well-defined generic planning workflow, and invest the next sprint in reproducibility and validation rather than adding more screens. This choice makes the project easier to defend in an evaluation because every claimed capability can be linked to source code, a test case and a visible limitation.

References and source notes

[1] AI Studio application URL
https://aakaash-adhinayak.ai.studio/
User-supplied hosted prototype link. Automated retrieval was unsuccessful in this review; screenshots and the exported source bundle were inspected instead.

[2] AviationWeather.gov Data API
https://aviationweather.gov/data/api/
Official public civil aviation weather data API documentation; connector status and terms must be verified before live claims.

[3] NASA Prognostics Center of Excellence Dataset Repository
https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/
Official NASA repository lists turbofan engine degradation simulation data and the C-MAPSS benchmark. It is not military telemetry.

[4] Airports Authority of India - Madurai Airport contacts
https://www.aai.aero/en/airports/contact-us/madurai
Official public contact details for the Airport Director office and terminal manager; proposed consultation target only, not evidence of a visit.

[5] India Meteorological Department - RMC Chennai
https://mausam.imd.gov.in/chennai/
Official regional weather observations, nowcasts, forecasts and public contact details; proposed consultation target only.